

User Guide

Assessment: Database Systems Assignment– ISYS2014/ISYS5008

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Introduction

This user guide explains how to create, populate, and use the Oscars Film Awards Database (2020–2024

All examples are based on **MySQL 8.0.43** on **Ubuntu 24.04 LTS**, using .sql files.

System Requirements

Software Needed

- MySQL Server 8.0+
- MySQL Client
- Python 3.10 or later
- mysql-connector-python module

3. Creating the Database

1. Open terminal
2. Login to MySQL:

```
mysql -u root -p
```

3. Create database:

```
CREATE DATABASE Nadim_22866899;
```

```
USE Nadim_22866899;
```

4. Creating All Tables (Implementation.sql)

Run your table-creation script here:

```
\. /home/Your_Ubuntu_Username/Desktop/Implementation.sql
```

This script creates:

- LOCATION
- CEREMONY
- CATEGORY
- FILM
- NOMINATION
- NOMINEE

With full PKs, FKs, and constraints

(From file: **Implementation.sql**)

5. Loading the Raw Data (CSV)

5.1 Enable local infile

```
mysql --local-infile=1 -u root -p
```

5.2 Run your Loading_Sample_data.sql

```
\. /home/ Your_Ubuntu_Username /Desktop/Loading_Sample_data.sql
```

This script performs:

Creates **OSCAR_RAW** temp table

Loads CSV using LOAD DATA LOCAL INFILE

Inserts Location data (Dolby Theatre, Union Station)

Inserts Ceremony rows (2020–2024)

Manually updates CeremonyDates from Wikipedia

Inserts Categories

Inserts Films

Inserts Nominations with cleaned winner values

Inserts Nominees (weak entity)

Updates FilmAge derived attribute

(All from **Loading_Sample_data.sql**)

6. Running SQL Queries (Part 3)

To run all 10 queries:

\. /home/ Your_Ubunutu_Username /Desktop/sql_queries.sql

This file contains:

- Level 1 queries (string functions, date functions, conditions)
- Level 2 queries (JOIN, GROUP BY, ORDER BY, aggregates)

(Source: **sql_queries.sql**)

7. Advanced Features (Stored Procedures, Trigger, View, Index)

Run your advanced implementation script:

\. /home/ Your_Ubunutu_Username /Desktop/Advanced_Implementation.sql

This file creates:

Stored Procedure 1 – AddNomination

Prevents duplicate nominations

CALL AddNomination(2023, 'DIRECTING', 'Barbie', 0);

Stored Procedure 2 – GetWinnersByYear

CALL GetWinnersByYear(2023);

Trigger – before_nomination_insert

Automatically calculates FilmAge

(Tested using INSERT INTO NOMINATION (...) VALUES (...);)

View – View_Winners

Lists all nominations where IsWinner = 1

Index

CREATE INDEX idx_categoryName ON NOMINATION(categoryName);

(From **Advanced_Implementation.sql**)

8. Python Connectivity (con1.py)

Install connector:

pip install mysql-connector-python

Create file:

nano con1.py

Paste your script (from **Python implementation.sql**)

Run the Python script:

python3 con1.py

The script performs:

- Connects to MySQL
- Displays 5 films
- Shows nominations for “Oppenheimer”
- Inserts a test film + nomination
- Shows inserted row
- Closes connection

9. How to Use the Database After Setup

View All Tables

SHOW TABLES;

Describe a Table

DESC NOMINATION;

Run Any Query

SELECT * FROM Category LIMIT 10;

Call Stored Procedure

CALL GetWinnersByYear(2024);

Insert New Nomination

CALL AddNomination(2024, 'BEST PICTURE', 'Python Demo Film', 0);

10. Troubleshooting

Error: LOCAL INFILE disabled

Run MySQL with:

```
mysql --local-infile=1 -u root -p
```

11. Conclusion

By following this guide and executing the .sql files in order, any user can fully recreate your Oscars Database, run queries, use advanced SQL features, and connect the database to Python for external access.